

# Theorem Examples

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This chapter demonstrates all the theorem-like environments available through the `callouty-theorem` and `custom-callout` extensions.

## Theorems

**Theorem 0.1** (Pythagorean Theorem). *In a right triangle with legs of length  $a$  and  $b$  and hypotenuse of length  $c$ , we have:*

$$a^2 + b^2 = c^2$$

See Theorem [0.1](#) for the famous Pythagorean theorem.

## Lemmas

**Lemma 0.1** (Fundamental Lemma). *Every non-empty subset of  $\mathbb{N}$  has a smallest element.*

The result in Lemma [0.1](#) is used to prove many theorems in number theory.

## Corollaries

**Corollary 0.1** (Rational Root Test Corollary). *If  $p/q$  is a rational root of the polynomial  $a_n x^n + \cdots + a_1 x + a_0$  with integer coefficients, then  $p$  divides  $a_0$  and  $q$  divides  $a_n$ .*

Corollary [0.1](#) follows directly from the Rational Root Theorem.

## Propositions

**Proposition 0.1** (Triangle Inequality). *For any real numbers  $x$  and  $y$ :*

$$|x + y| \leq |x| + |y|$$

The triangle inequality (Proposition 0.1) is fundamental in analysis.

## Conjectures

**Conjecture 0.1** (Goldbach's Conjecture). *Every even integer greater than 2 can be expressed as the sum of two prime numbers.*

Conjecture 0.1 remains one of the oldest unsolved problems in number theory.

## Definitions

**Definition 0.1** (Derivative). The derivative of a function  $f$  at a point  $x$  is defined as:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided this limit exists.

See Definition 0.1 for the formal definition of a derivative.

## Examples

**Example 0.1** (Solving a Quadratic Equation). Solve  $x^2 - 5x + 6 = 0$ .  
We can factor this as  $(x - 2)(x - 3) = 0$ , so the solutions are  $x = 2$  and  $x = 3$ .

Example 0.1 shows how to solve a simple quadratic equation.

## Exercises

**Exercise 0.1** (Integration Practice). Compute the following integral:

$$\int_0^1 x^2 dx$$

Try Exercise [0.1](#) to practice basic integration.

## Proofs

**i** Proof (Proof of the Pythagorean Theorem)

*Proof of the Pythagorean Theorem.* Consider a square with side length  $a + b$ . We can divide this square into four right triangles and a central square with side length  $c$ . The area equations show:

$$a^2 + b^2 = c^2$$

□

**i** Proof (Proof that the derivative of a constant is zero)

*Proof that the derivative of a constant is zero.* Let  $f(x) = c$  for some constant  $c$ . Then  $f'(x) = 0$ . □

## Solutions

Solution (Solution to Exercise on Integration)

*Solution 0.1* (Solution to Exercise on Integration). We use the power rule for integration:

$$\int_0^1 x^2 dx = \left[ \frac{x^3}{3} \right]_0^1 = \frac{1}{3}$$

#### Solution (Alternative Solution to Quadratic Equation)

*Solution* (Alternative Solution to Quadratic Equation). For the quadratic equation, we can use the quadratic formula:

$$x = \frac{5 \pm 1}{2}$$

This gives us  $x = 3$  or  $x = 2$ .

## Remarks

*Remark 0.1* (Convergence Note). The limit in Definition 0.1 may not exist for all functions at all points. For example, the absolute value function  $|x|$  is not differentiable at  $x = 0$ .

Remark 0.1 highlights an important limitation of the derivative definition.

## Mixed Example with Theorem and Solution

Here's an example that combines multiple theorem types:

**Theorem 0.1** (Mean Value Theorem). *If  $f$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ , then there exists a point  $c \in (a, b)$  such that:*

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

**Exercise 0.1** (Applying the Mean Value Theorem). Let  $f(x) = x^2$  on  $[0, 2]$ . Find the value of  $c$  guaranteed by the Mean Value Theorem (Theorem 0.1).

#### Solution (Solution to Mean Value Theorem Exercise)

*Solution 0.1* (Solution to Mean Value Theorem Exercise). We have  $f(x) = x^2$ , so  $f'(x) = 2x$ .

By the Mean Value Theorem:

$$f'(c) = \frac{f(2) - f(0)}{2 - 0} = \frac{4 - 0}{2} = 2$$

Since  $f'(c) = 2c$ , we need:

$$2c = 2$$

$$c = 1$$

Therefore,  $c = 1$  is the value guaranteed by the Mean Value Theorem.

## References